Shape and Scale: What Decides Success for BO and CMA-ES?

Isaac Soong Hong Wei (A0272752B) Paul Ng Yin Chen (A0272249Y)

GitHub: https://github.com/isaacsoong055/DSA4212

Abstract

Bayesian optimisation (BO) and CMA-ES both carry an implicit model of the landscape: CMA-ES a Gaussian search distribution with a covariance (its shape) and a step size (its scale), BO a Gaussian-process surrogate whose kernel metric and lengthscales play the same two roles. We test one explanation in two settings: a method succeeds when its model's shape and scale can match the problem, whatever its family. *Part A (multimodality)* asks whether the model's scale lets a method *find* and then *refine* a narrow distant basin: neither default method finds it; CMA-ES's step size cannot do both jobs at once, and BO succeeds once its lengthscale is capped (1/50 to 44/50 on fresh seeds) while acquisition tuning helps far less. *Part B (geometry)* asks whether the model can *tilt*: on rotated, ill-conditioned valleys diagonal CMA-ES, the ARD kernel and TuRBO's axis-aligned trust region all fail, while their tilting counterparts do not, once BO's own fitting failures are removed. In both parts the decisive BO changes were to the model fit, not the acquisition function. All results are in 2-D and pre-registered predictions are reported whether they held or failed.

1 Introduction

1.1 Motivation and the two methods

Many problems give only function values (tuning hyperparameters, calibrating a simulator), with no gradients and costly evaluations, so the budget is counted in evaluations. Bayesian optimisation (BO) and the covariance matrix adaptation evolution strategy (CMA-ES) are two of the most widely used methods here, with opposite designs: BO models the function, CMA-ES adapts a search distribution.

BO [5] fits a Gaussian-process (GP) surrogate to all evaluations so far. Its posterior gives a prediction and an uncertainty everywhere, and an acquisition function, such as expected improvement (EI), turns these into the next point. CMA-ES [7] samples a population of λ candidates from a Gaussian $\mathcal{N}(m, \sigma^2 C)$, ranks them, moves the mean towards the best, and adapts the covariance C and step size σ from successful steps, using only the ranking of the values. BO is sample-efficient but costly per step and depends on its model being right; CMA-ES needs more evaluations but is nearly free per step.

1.2 What the literature says, and what it leaves open

Hansen's tutorial [7] states the invariances CMA-ES is designed to have: to any order-preserving transformation of f, and to rotations and other linear maps of the search space provided the initial conditions are mapped too. Restart schemes that grow or mix the population size (IPOP, BIPOP [2, 6]) target multimodal functions. TuRBO [4] restricts BO to local trust regions whose sides are set per coordinate from the ARD lengthscales, an axis-aligned shape.

BO's results also depend on rarely reported settings. Ament et al. [1] show that EI's values and gradients vanish numerically, so the acquisition optimiser can fail, and Siivola et al. [16] show that GP-based BO over-explores the boundary of the search box. A comparison run with library defaults therefore mixes the family with hidden choices (kernel fitting, lengthscale bounds, noise handling, step size); which of these actually decides success is open.

1.3 Problem formulation

Landscapes differ in modality (one basin or many, wide or narrow) and geometry (well conditioned or stretched along a rotated valley), while BO and CMA-ES each have several tunable settings. We ask how adjusting those settings changes how well each method finds the optimum: which property of a method decides the outcome, and does it matter more than the choice of family?

Guiding hypothesis. Each method carries an implicit model of the landscape: what it assumes the function looks like when choosing the next point. The model has a *shape* (the orientation and relative stretch of the directions it treats as different) and a *scale* (the distance over which it treats the function as changing). For CMA-ES the model is the search distribution: the covariance C sets the shape and the step size σ the scale, i.e. how far from the mean it samples. In BO, this role is played by the GP's kernel. Its metric sets the shape of similarity: a full metric can rotate and mix input directions, while a diagonal ARD metric only rescales each axis independently. The lengthscale sets the scale, meaning the distance over which two points' function values remain strongly correlated. BO also has the acquisition function (its form, exploration weight and optimiser), which chooses the next point given the model and is tested as a competing explanation. A method succeeds when its model can match the landscape's shape (orientation, conditioning) and scale (the size of the features to find and resolve), and fails when a restriction or fitting failure prevents it, whichever its family.

1.4 Shared experimental approach

Both parts count the budget in objective evaluations, initial designs and restarts included; run every variant on the same seeds; change one setting at a time against a baseline; and commit predictions before the runs that test them,

reporting each as held, partly held or failed. A method's recommendation is its best evaluated point; in Part B's noisy experiments it is chosen by a stated rule (lowest reading, model estimate or averaged repeats). Every recommendation is scored on the true, noise-free objective. The parts use different budgets, measures and BO code (BoTorch [3]; NumPy/SciPy), so their numbers are never compared directly.

1.5 The two directions

Part A, multimodality, asks whether each method can find a narrow, distant global basin and then refine it, within 300 evaluations. It varies CMA-ES's population size, restarts and step size, and BO's acquisition function, GP fitting settings and kernel, on a decoy bowl with a narrow Gaussian well, with a sphere control and Rastrigin and Schwefel for validation.

Part B, geometry and parameterisation, asks whether a method's model can tilt to a rotated, ill-conditioned valley, and which of the invariances in Section 1.2 (to monotone transformations of f and linear maps of the space) hold in practice. It varies conditioning, rotation and whether the model can rotate (full against diagonal covariance; full against ARD metric; rotated against axis-aligned trust region), and tests units, monotone transformations, the box boundary and noise, on ellipsoids, Rosenbrock (COCO [8]) and a noisy quadratic, mostly within 200 evaluations.

2 Part A (Multimodality): Can CMA-ES and Bayesian optimisation find a narrow, distant global basin?

Summary. On a landscape with a wide decoy basin and a narrow, deep global well, neither baseline optimiser reliably finds the well within $N = 300$ evaluations (Exp 0: CMA-ES 0 of 50 runs succeed; Bayesian optimisation, BO, 5 of 50). Success was split into *discovery* (a sample lands in the well) and *refinement* (the well is then exploited). For CMA-ES, a larger population and restarts improve discovery but not success, because refinement is limited by the initial step size. For BO, bounding the surrogate's lengthscale raises success from 5/50 to 39/50 (44/50 on fresh seeds); the gain does not carry over to Rastrigin and Schwefel, where the baseline lengthscale never reaches the bound.

2.1 Problem and experimental set-up

2.1.1 The test landscape

All method comparisons use one tunable function on $[-5, 5]^2$: a wide, shallow *decoy* bowl with a narrow, deep Gaussian *well* cut into it,

$$f(x) = \frac{\|x - c\|^2}{s} - A \exp\Big(- \frac{\|x - x_g\|^2}{2w^2} \Big), \qquad c = (0, 0),\ s = 10,\ A = 4. \tag{1}$$

The decoy centre c has $f = 0$. The well, of width w (default 0.3), is centred at $x_g = (3, 3)$ (**far**) or $(2, 0)$ (**near**); the global minimum is $f^* = -2.208$ (Figure 1). The bowl slopes towards c everywhere, so nothing local points to the well. Its width w (how *narrow*) and position (how *distant*) are the two knobs of the study. All distances and widths in this report are in the units of the search box $[-5, 5]^2$, which is 10 wide; for example the far well is $\|(3, 3)\| = 4.24$ from the decoy centre and the near well is 2 from it.

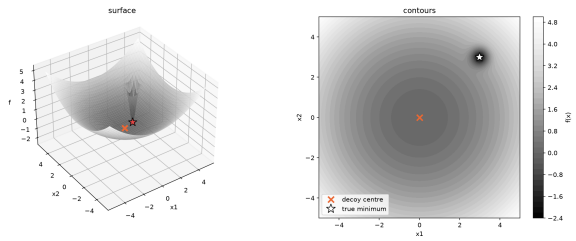


Figure 1. Primary landscape (far well, $w = 0.3$): surface (left) and contours (right).

2.1.2 Success criterion

A run's *regret* is $f(x_{\text{best}}) - f^*$ for the best point it evaluated. A run **succeeds** if its regret is at most $\varepsilon = 0.05$; otherwise it **fails**, almost always by converging on the decoy (regret about 2.21, the well's depth). Success needs two stages, kept apart throughout: **discovery**, an evaluation within w of the well centre (a *touch*), and **refinement**, going on from that touch to regret $\leq \varepsilon$. The threshold ε was chosen from random sampling alone: the largest value for which random search with $N = 300$ succeeds in at most 5% of runs. A count $k/50$ is given with a 95% Wilson interval [14] $[\text{low}, \text{high}]$, the range of true success probabilities consistent with k successes in 50 runs; methods are called different only when these intervals, or that of their difference [13], separate.

Table 1. Baseline settings and the experiments that vary them. "Library" = software default; "chosen" = value fixed before the large runs. BO on the sphere, Rastrigin and Schwefel uses 20 seeds.

Setting (meaning)	Baseline (source)	Varied in
CMA-ES (`cma` library); Exp 0 = default CMA-ES		
population λ (points per generation)	6 (library)	Exp 1: 12, 24
restarts (fresh start after convergence)	none	Exp 2: IPOP, BIPOP
step size σ_0 (initial spread; box width 10)	2 (chosen)	Exp 3: 0.3, 0.5, 1, 4
start x_0 (initial mean)	$U[-4, 4]^2$ (chosen)	Exp 3: "oracle" start, mean at distance 1.0 from the well centre
BO (BoTorch); Exp 0 = default BO		
acquisition (next-point rule; ξ = exploration bonus)	EI, $\xi = 0$ (standard)	Exps 4, 5: ξ up to 2; LCB; Thompson sampling
lengthscale bound (allowed GP lengthscale range)	$[0.05, 20]$ (chosen)	Exp 6: upper bound 5, 2, 1, 0.5, 0.25
noise (treats f as noise-free)	fixed 10^{-6} (chosen)	Exp 6: 10^{-3} to 1; learned; lengthscale prior
kernel (GP smoothness)	Matérn-5/2 (BoTorch)	Exp 7: Matérn-3/2, 1/2, RBF (at upper bound 20 and 1)
initial design, refit	10 random points; refit every 5 (chosen)	not varied
Landscape (both methods)		
well (narrowness, distance)	$w = 0.3$, far (chosen)	Exp 8: $w \in \{0.1, 0.3, 0.6, 1.0\}$, far and near
function	primary landscape	Exp 9: sphere; Exp 10: Rastrigin, Schwefel

2.1.3 Methods, settings, control and transfer functions

Table 1 gives the baseline of every setting (library default or a value frozen before the large runs) and the experiment that varies it. Exp 0 (the baselines) is the starting point; each later experiment follows from the earlier results and changes exactly one row. Restart triggers were chosen by how often they fire, never by success. The **sphere** (single basin) is the control: a setting that helps there as well does not owe its gain to multimodality. **Rastrigin** and **Schwefel** are *transfer tests* of whether conclusions carry over to standard functions.

2.1.4 Fair comparison

Every run uses exactly $N = 300$ evaluations, including restarts and BO's initial design (enforced in code), and each experiment uses 50 seeds $(1000, \ldots, 1049)$ on the far well with $w = 0.3$, unless marked (the $N = 1500$ CMA-ES check, oracle starts, fresh seeds, BO on 20 seeds); these are never compared with BO. One factor changes at a time, on the same seeds. The best variant of each method is re-run on *fresh* seeds $2000, \ldots, 2049$, so selecting the best of several variants cannot inflate the headline. Baselines were frozen, and the hypotheses committed to the repository (`predictions.md`, 22 October 2026), before any success rate was seen; later explanations are marked as hypotheses or *post hoc*.

2.2 Results and analysis

2.2.1 Exp 0: how often do the defaults find the well, and where do they go?

The intervals overlap, so the methods cannot be told apart at 50 seeds. Almost all runs of both methods end at regret 2.21: they converged on the decoy.

Table 2. Exp 0 (far well, $w = 0.3$, 50 seeds). A touch = an evaluation within w of the well centre.

Method	Successes	95% interval	Discovery: runs touching the well	Refinement: succ. given a touch	Median final regret
Default CMA-ES	0/50	$[0.00, 0.07]$	1	0/1	2.21
Default BO (EI, $\xi = 0$)	5/50	$[0.04, 0.21]$	5	5/5	2.21
Random sampling	2/50	$[0.01, 0.13]$	–	–	1.11

CMA-ES moves its sampling mean towards the best-ranked points, and step-size adaptation shrinks the spread σ once the search has converged [7]. In the bowl the best points always lie towards the centre, so σ shrinks from 2 to about 10^{-3} and from evaluation 100 on at least 97.5% of samples lie within distance 0.5 of the decoy centre, while the well centre is at distance 4.24 from it. The one run that touched the well did not refine it: the many decoy samples outweighed the single well sample in the update (Exp 3).

BO fits a Gaussian process (GP) and evaluates next where expected improvement (EI) is largest [5, 11]. The GP's *lengthscale* is how far apart two points must be before the model treats their values as unrelated. In 42 of 50 runs it reached its upper bound of 20, and in all 45 failures no sample entered the well: the model sees one smooth bowl and keeps sampling its centre. So *default BO fails at discovery, not refinement*: all 5 runs that touched the well succeeded (Table 2). **Why the maximum lengthscale?** (hypothesis): maximum-likelihood fitting favours smooth

explanations [15], and with noise-free data the fitted lengthscale is unstable, becoming infinite when the data are flat [12]; we suggest that data which look like one smooth bowl push it towards that limit. Here the noise is fixed at 10^{-6} and the data are an exact bowl.

2.2.2 Exps 1–3: CMA-ES

Observation: in Exp 0, CMA-ES converged on the decoy; only 1 of 50 runs touched the well, and it did not refine it. Hypotheses A and B act on discovery, C on refinement.

Hypothesis A: a larger population finds the well more often and succeeds more (Exp 1). More points per generation should put more early samples in the well [7]. *Test:* $\lambda \in \{6, 12, 24\}$. *Result* (Table 3): discovery improved (touches: 1, 7, 13 of 50 runs) but refinement did not (successes given a touch: 0/1, 0/7, 0/13). Since $\lambda = 24$ allows only 13 generations, a follow-up experiment with $N = 1500$ checked whether the failure was caused by this cut-off; it again gave 0/50. *Answer:* a larger population improves discovery, not success. *Why (tested in Exp 3):* a larger λ adds samples but leaves the spread $\sigma_0 = 2 \gg w$ unchanged, so the many decoy samples still outweigh a well sample in the update.

Hypothesis B: restarts help (Exp 2). Restarting after convergence, with larger populations (IPOP [2]) or a mix of large and small (BIPOP [6]), gives fresh chances to land in the well. *Result:* 0/50 for none, IPOP and BIPOP (identical here: only one restart fits in 300 evaluations); the well was touched in 6 runs, like a fresh $\lambda = 12$ start (7). *Answer:* no; a restart is a new attempt with the same small chance.

Table 3. Exps 1, 2: every setting gives 0/50 successes (interval $[0.00, 0.07]$).

Setting	Successes	Runs with a sample in the well	Median generations
$\lambda = 6$ (default)	0/50	1	50
$\lambda = 12$	0/50	7	25
$\lambda = 24$	0/50	13	13
IPOP / BIPOP (identical)	0/50	6	–

Hypothesis C: the step size is too large to refine a narrow well (Exp 3). With $\sigma_0 = 2 \gg w = 0.3$, the many selected decoy points can outweigh one well sample. *Test:* vary σ_0 from the usual random start (success = finding *and* refining) and from an "oracle" start whose initial mean is at distance 1.0 from the well centre (success = refining alone). *Result* (Table 4): from the random start no setting exceeds 1/50; from the oracle start success falls from 33/50 at $\sigma_0 = 0.3$ to 2/50 at the default 2. With $\sigma_0 = 2$ the search mean drifts within about five generations to the decoy centre; with $\sigma_0 = 0.3$ it converges onto the well. Finding and refining pull σ_0 in opposite directions: a small σ_0 refines a nearby well but never reaches a distant one; a large one touches it more but cannot exploit it. *Answer:* yes; this is why λ and restarts, which only affect touching, left success unchanged.

Follow-up to Exp 3 (does only the ratio σ_0/w matter?). Here the initial step size is $\sigma_0 = kw$, a multiple $k = 1, 3, 7, 20$ of the well width, tested at $w = 0.1$ to 1.0, from an oracle start at distance $10w/3$ from the well centre (50 seeds each). Narrow wells repeat Exp 3 (at $w = 0.1$: 44/50 for $k = 1$, 0/50 for $k = 20$), but at $w = 1.0$ it reverses: 24/50 for $k = 1$, 40/50 for $k = 20$. Success is therefore not a function of σ_0/w alone; a wide well is found by a broad cloud. The `cma` library caps the step size at about 4 in this box, so $k = 7$ and 20 coincide for $w \geq 0.6$; Exp 3 ($\sigma_0 \leq 4$) is unaffected.

Table 4. Exp 3: successes out of 50 (touches in brackets).

σ_0	0.3	0.5	1	2 (default)	4
Random start	0 (0)	0 (0)	1 (2)	0 (1)	0 (4)
Oracle start	33 (35)	31 (39)	10 (23)	2 (11)	1 (11)

2.2.3 Exps 4–7: BO

Observation: default BO fails at discovery.

Hypothesis D: more exploration helps (Exps 4, 5). *Test:* raise EI's exploration parameter ξ (a bonus on uncertain points); replace EI by the lower confidence bound (LCB) or Thompson sampling. *Result* (successes out of 50): moderate exploration helps a little ($\xi = 0.1$: 17/50, +0.24, interval $[+0.08, +0.39]$), too much hurts ($\xi = 0.01, 0.5, 1, 2$: 8, 12, 5, 3), and LCB (3–4/50) and Thompson sampling (3/50) do not help. With large ξ the well is found late and successes given a touch fall from 17/45 at $\xi = 0.1$ to 3/42 at $\xi = 2$; why late finds are exploited poorly is unresolved. *Answer:* more exploration helps only modestly: the best setting ($\xi = 0.1$) lifts success from 5/50 to 17/50, and further increases make it worse.

Hypothesis E: the fitted lengthscale is the limit (Exp 6). Exploration tuning gave at most 17/50, while the lengthscale sat at its bound in 42/50 runs. *Test:* lower the upper bound to 5, 2, 1, 0.5, 0.25 (one value per arm); separately, change the noise level and add BoTorch's lengthscale prior. *Result* (Table 5): a bound of 1 (about $3w$) gives 39/50, +0.68 over the default (interval $[+0.51, +0.79]$), the most effective change in the study. A smaller bound

makes distant points unrelated, so EI keeps sampling unexplored regions (about 100 distinct 1×1 cells with bound 1, against 10) and a sampled well is exploited. A bound of 0.25 ($\approx w$), noise settings and a soft prior do not help (at most 6/50). *Answer:* yes; a hard bound on the lengthscale is what matters.

Table 5. Exp 6: upper bound on the BO lengthscale; difference vs the default (bound 20).

Bound	20 (default)	5	2	1	0.5	0.25
Successes	5/50	11/50	28/50	**39/50**	22/50	6/50
95% interval	[0.04, 0.21]	[0.13, 0.35]	[0.42, 0.69]	[0.65, 0.87]	[0.31, 0.58]	[0.06, 0.24]
Difference	–	+0.12	+0.46	+0.68	+0.34	+0.02
Runs touching the well	5	12	29	41	24	6

Hypothesis F: the gain depends on the kernel (Exp 7). With the lengthscale upper bound at 20 no kernel differs from Matérn-5/2 (5/50). With the upper bound at 1 the rougher Matérn-3/2 and 1/2 drop to 6/50 and 3/50, while RBF matches Matérn-5/2 (39/50). A likely reason (hypothesis, from the kernel formulas [15]): the same lengthscale gives a different correlation range. At a distance equal to the lengthscale the correlation is 0.37 (Matérn-1/2), 0.48 (3/2), 0.52 (5/2) and 0.61 (RBF), so with the bound at 1 the rough kernels behave as if it were tighter, and they rarely touch the well (6 and 3 runs, against 41). Testing this needs a separate bound per kernel. *Answer:* yes, for the rougher kernels.

2.2.4 Exp 8: headline result across well widths

Test: $w \in \{0.1, 0.3, 0.6, 1.0\}$ for the far well and $w \in \{0.1, 0.3\}$ for the near well (at $w \geq 0.6$ it merges with the bowl). Defaults use seeds 1000–1049; the best variant of each method (CMA-ES $\sigma_0 = 1$, chosen on 1/50; BO bound 1) is re-run on fresh seeds (Figure 2).

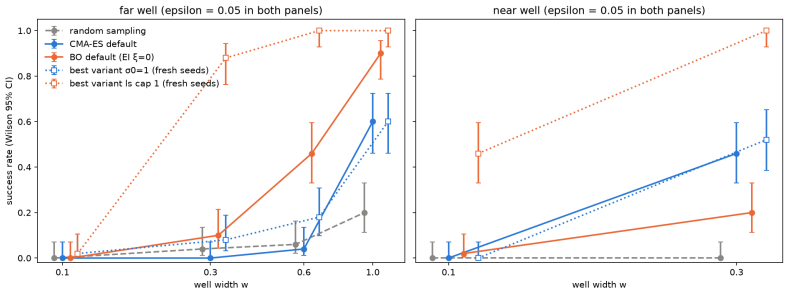


Figure 2. Exp 8: success rate against w for the far well (left) and the near well (right); Wilson 95% intervals, successes out of 50 printed. Solid: defaults (seeds 1000–1049); dotted: best variants (fresh seeds 2000–2049).

Width. Success rises with w for every method (default CMA-ES: 0/50 at $w = 0.3$, 30/50 at $w = 1.0$). The main reason is the size of the target: the well's catchment is a disc of radius w, so 300 random points in the 10×10 box touch it about $300\pi w^2/100 = 0.1$, 0.8, 3.4, 9.4 times for $w = 0.1$, 0.3, 0.6, 1.0, matching random search (0, 2, 3, 10 of 50 successes); a wide well is also easier for a broad cloud or a GP to notice. BO with bound 1 reaches 44/50 at $w = 0.3$ on fresh seeds (39/50 on the selection seeds) and 50/50 at $w = 0.6$ and 1.0, but only 1/50 at $w = 0.1$: even the bounded model rarely samples so narrow a well. CMA-ES with $\sigma_0 = 1$ shows no gain at any width.

Distance. On the *near* well, default CMA-ES beats default BO at $w = 0.3$ (23/50 and 30/50 on the two seed sets, against 10/50 and 13/50), whereas on the far well it does not. A likely reason (hypothesis): the near well lies at distance 2 from the decoy centre (the far well: 4.24), so CMA-ES's initial cloud (spread $\sigma_0 = 2$) and the mean's path to the centre pass close to it. Unexplained: default BO at $w = 0.6$ scores 23/50 on seeds 1000–1049 but 10/50 on fresh seeds.

2.2.5 Exps 9–10: control and transfer

Sphere (control). Every setting succeeds in every run: the sphere has one basin, and the success threshold $\varepsilon = 0.05$ is far above every final regret. The settings differ only in *precision*: a larger population gives a *worse* final regret (median 1.6×10^{-10}, 9.8×10^{-8}, 6.0×10^{-5} for $\lambda = 6, 12, 24$). The cause is the fixed budget: on a single smooth basin CMA-ES shrinks its regret by a roughly constant factor per generation, so precision is set by the number of generations, and $N = 300$ allows only 50, 25 and 13 for $\lambda = 6, 12, 24$. A bigger population takes a slightly better step (it averages over more points), but that does not make up for the lost steps.

Rastrigin and Schwefel (transfer). The bound's large gain does *not* carry over cleanly (Table 6). Success rises with the bound on both functions, +0.10 on Rastrigin (interval $[-0.19, +0.37]$) and +0.20 on Schwefel ($[-0.0003, +0.42]$), but with 20 seeds neither can be separated from zero, and BO already scores 16/20 on Schwefel, so 20/20 is the ceiling. CMA-ES population size has no distinguishable effect. The logged fits show where the bound can act (post hoc; last column of Table 6). On the sphere and the primary landscape it is 8–9 after the initial design and ends at the bound

of 20 (20 of 20 and 86 of 100 runs). On Rastrigin it falls from 1.3 to 0.41, shorter than the function's period of 1: it ends below 1 in all 20 runs, so a bound of 1 can only act early. On Schwefel the median rises from 0.56 to 1.28 and ends above 1 in 14 of 20 runs, so a bound of 1 *is* active there, which fits its small increase. Neither function reaches the bound of 20. A likely reason (hypothesis): the sphere and primary landscape are a smooth bowl, which a long lengthscale fits best, whereas Rastrigin and Schwefel oscillate over distances of about 1. This gives a rule: *the large gain needs the baseline lengthscale to run to its maximum and a narrow basin to be hidden.* The primary landscape meets both; the sphere only the first (and the bound costs accuracy there); Rastrigin and Schwefel neither, so the bound is inactive or mildly active and the gain small and uncertain.

What follows. A bound should not be applied blindly: first check whether the fitted lengthscale sits at its upper limit, then bound it, at a few well widths. Confirming the rule needs a per-function bound (e.g. below 0.4 on Rastrigin) or more seeds. BO also beats CMA-ES on Schwefel (16/20 against 2–8 of 50), unexpectedly; no bug or configuration slip was found.

Table 6. Exp 10: successes (CMA-ES, random: 50 seeds; BO: 20); $\varepsilon = 0.2$ (Rastrigin), 5.0 (Schwefel). Last column: median fitted lengthscale of default BO (box units, upper bound 20) after the initial design → at the end of the run (logged fits).

	CMA-ES $\lambda = 6$	$\lambda = 12$	$\lambda = 24$	$\sigma_0 = 1$	BO default	BO bound 1	Random	BO default lengthscale
Rastrigin	8/50	11/50	4/50	7/50	8/20	10/20	1/50	$1.29 \to 0.41$
Schwefel	8/50	6/50	2/50	4/50	16/20	20/20	2/50	$0.56 \to 1.28$

2.3 Limitations

- **Scope and selection.** One main landscape in two dimensions with one ε; the Rastrigin and Schwefel explanation rests on four landscapes and 20 BO seeds, too few to separate +0.20 on Schwefel from zero. Bound 1 is the best of five values, so 39/50 is optimistic (44/50 on fresh seeds); 50 seeds give intervals of about ±14 points.
- **Frozen settings and mechanisms.** The BO failure depends on the chosen bound of 20; the explanations for the maximal lengthscale, late finds, kernels and the near well are hypotheses.

2.4 Conclusion

At a fixed budget of 300 evaluations, both baselines fail on a narrow, distant well because they never *find* it. For CMA-ES, a larger population and restarts only add chances to touch it, and a small step refines only a nearby well. For BO, bounding the lengthscale at about three well widths turned 5/50 into 39/50, provided the baseline lengthscale runs to its upper limit.

3 Part B: can the search tilt? (geometry and parameterisation)

3.1 Question, set-up and predictions

Near a minimum $f \approx f^* + \frac{1}{2} h^\top H h$. The Hessian's eigenvalues are the curvatures; their ratio κ is the conditioning (a narrow valley when large); its eigenvectors are the valley's orientation, which a rotation changes without changing κ. A full covariance (CMA-ES) or a full kernel metric $M = LL^\top$ (BO) can align with a rotated valley; a diagonal covariance, the standard ARD kernel and TuRBO's axis-aligned trust region [4] cannot. We predicted that (i) each axis-restricted method pays a penalty under rotation that its tilting counterpart does not, (ii) only when the problem is both ill-conditioned and rotated, and (iii) the tilting version pays a small price when no tilt is needed.

2-D ellipsoids with log-spaced eigenvalues from 1 to κ, the optimum on a radius-2 sphere, aligned or Haar-rotated, 15 instances (COCO-style; seed = instance). CMA-ES follows Hansen's tutorial, with a diagonal ablation. BO is GP-EI (initial design of $2d + 4$, EI over 200 uniform candidates polished by L-BFGS-B, targets log-transformed) with a MAP lengthscale prior on the metric's eigen-lengthscales [10], so that the prior does not favour axis-aligned metrics. Both get the box $[-5, 5]^2$ (BO searches it; CMA-ES starts at its centre with $\sigma_0 = 3$) and 200 evaluations. We report regret after that fixed budget because BO cannot run much longer (each run takes minutes), and give the runtime to a target, the measure COCO recommends [9], where the two views differ.

Experiments (scripts `geometry/code/run_e*.py`): E1 rotation and conditioning sweep, CMA-ES; E2 BO against CMA-ES head to head; E2c the GP kernel's learned axis; E2e diagnosis of BO's stall; E3 invariances (monotone maps, units); E3b units check with local search; E4 optimum near the box edge; E4b why BO samples the edge; E5 Rosenbrock and a noisy quadratic at 200 evaluations; E6 noisy budget splits and recommendation rules; E6b the recommending GP; E7 CMA-ES restarts; E8 Matérn kernel; E9 Rosenbrock, long CMA-ES runs; E10 trust-region BO; E11 the boundary trap; E12 locally fitted trust region.

3.2 CMA-ES fails under rotation only when it cannot tilt (E1, E7)

Diagonal CMA-ES fails on a rotated, ill-conditioned valley and full CMA-ES does not, as predictions (i) and (ii) require. At $\kappa = 1{,}000$ diagonal CMA-ES solves every instance at 0° and 90° and none between 30° and 60° (0/15 at each, $[0, 0.20]$), while full CMA-ES solves every angle; at $\kappa = 100$ both succeed everywhere. The 90° control only swaps the axes and recovers fully, so the rotation, not the conditioning, causes the failure. Full CMA-ES is rotation-invariant

BO variant (2-D, 15 instances, 200 evaluations)	rotated $\kappa = 10^4$	aligned $\kappa = 10^4$
EI, squared exponential, noise fixed	$+1.40$ [$+0.68, +1.69$]	-0.13 [$-1.15, +0.19$]
EI, Matérn-5/2	$+1.06$ [$+0.59, +1.43$]	-0.13 [$-0.70, +0.54$]
EI + local candidates	$+1.28$ [$+0.12, +2.36$]	-2.96 [$-3.44, +2.61$]
Trust region, ARD box vs rotated box	$+2.92$ [$+1.27, +3.79$]	-0.39 [$-3.11, +4.65$]

Table 7. Part B. Axis-restricted minus tilting version, median $\log_{10} f$ at 200 evaluations, paired by instance (+ = axis-restricted worse). Wide aligned intervals come from the boundary trap.

in distribution (405 against 408 median evaluations under Haar rotation). IPOP-style restarts leave the diagonal failure in place (1/15 against 3/15 reaching 10^{-5} at rotated $\kappa = 10^4$; difference -0.13 [$-0.39, +0.13$], not significant), since a restarted diagonal run still cannot tilt. Both use 4,000 evaluations; E1 starts at $(2, \ldots, 2)$ with $\sigma_0 = 2$ and a target of 10^{-8}, E7 at the box centre with $\sigma_0 = 3$.

3.3 BO first showed no penalty, because a fitting failure stalled every run (E2, E2e)

BO's first head-to-head contradicted prediction (i), and the cause was a fitting failure, not the geometry. ARD's learned long axis is off by exactly 45° on 45° problems (E2c), while the full metric is within 0.2°–4°, so ARD cannot tilt; yet its rotation penalty was $+0.05$ [$-0.53, +0.63$] decades at $\kappa = 100$, while diagonal CMA-ES paid $+1.70$ [$+0.41, +2.57$]. BO was stalling near $10^{-2.6}$ even on the round bowl. Per-iteration logs ruled out the target transform's floor and the candidate search, and showed the GP fitting a noise term (SD 0.2–0.4) to a noise-free function: it explained its best points away as noise, its prediction at the incumbent sat 2–2.5 times the gap to the second-best point above the observed value, and EI underflowed below 10^{-12} in 97–100% of late iterations. BO fits the GP to $\log f$, because raw values on these problems span many orders of magnitude and would otherwise be dominated by the few largest. But the log turns the optimum into a sharp dip. The GP's lengthscales (one per direction, with ARD) are the same everywhere in the box, so following the dip would make its whole surface wiggly; since it also learns its noise level, it finds it cheaper to call the dip noise: the marginal likelihood favours the simpler explanation [15, Ch. 5]. Each of the three is needed: fixing the noise removes the way out (the fix below), dropping the log removes the dip, and a kernel whose lengthscales vary with location (non-stationary) would let the GP bend only near the optimum. We tested only the first; a locally fitted GP, a crude form of the third, lost for lack of data (E12, Section 3.5). Fixing the noise variance at 10^{-6} lifted BO by -1.17 [$-1.71, -0.74$] decades, whereas LogEI [1], which fixes the underflow but not its cause, changed nothing ($+0.01$ [$-0.79, +0.47$]). For a deterministic objective the noise should be fixed near zero; we had kept the usual default (as in BoTorch) of learning it. Section 3.8 tests the fix under noise.

3.4 Once BO works, the axis-restricted version pays in every variant (E2, E8, E10)

With the fix, BO-ARD pays a rotation penalty and BO-full does not, as predictions (i) and (ii) require, across two kernels, two acquisition searches and a trust region. Run on one machine, BO-ARD's penalty became $+2.01$ [$+1.28, +2.97$] decades at $\kappa = 10^4$ and BO-full's stayed non-significant (Fig. 3). Table 7 repeats the comparison with a Matérn-5/2 kernel, 200 extra candidates around the incumbent, and a TuRBO-style trust region [4] whose box is either axis-aligned with ARD-scaled sides or rotated to the full metric; in every variant the axis-restricted version ends significantly behind on rotated problems. Where no tilt is needed, tilting costs a little, as prediction (iii) expects: on aligned $\kappa = 100$ the axis-aligned trust region is 0.57 decades better (-0.57 [$-1.04, -0.27$]). The reason is shared: the ARD kernel and the published trust region both scale each coordinate separately, so neither can align with a rotated valley. Measured by runtime to a target, BO's standing depends on its search: on the round bowl CMA-ES reaches 10^{-5} in every run (median 142–154 evaluations), BO with the noise fix in only 0–4 of 15 runs (at most $[0.11, 0.52]$) within 200, and BO with local candidates in every run (median 41–56). After the fix, plain EI stalls near 10^{-4} for the reason ruled out before it: its 200 uniform candidates rarely land in the small region below the incumbent (in 98% of late iterations a search near the incumbent found over ten times the EI of the chosen point), and local candidates remove this. Across families, the trust region suited to each problem beats full CMA-ES at 200 evaluations: the axis-aligned box on aligned $\kappa = 100$ (-1.36 [$-2.02, -0.85$]) and the rotated box on rotated $\kappa = 10^4$ (-3.32 [$-4.24, -1.82$]; the axis-aligned box there: -0.41 [$-1.24, +0.24$]), while with plain EI CMA-ES wins by 200 evaluations at $\kappa \leq 100$.

3.5 On a bending valley, local orientation and local search decide (E9, E12)

On Rosenbrock (COCO f8, f9 [8]), whose valley bends, what matters is the local orientation. Even "aligned" Rosenbrock is tilted 63.5° near its optimum (local $\kappa \approx 2{,}500$), so with 4,000 evaluations full CMA-ES reached 10^{-5} in every run (median about 500 evaluations) but diagonal CMA-ES in only 0/15 (f8; $[0, 0.20]$) and 4/15 (f9) runs, against our prediction that both would. For BO at 300 evaluations, the trust region beat plain BO by 3.5–3.7 decades and its tilted box beat the axis-aligned one by 2.0–2.5 decades, even on f8: plain BO's stationary kernel can only average the directions along a curving valley, while inside a small box the valley is nearly straight. Part of this lead may come from the trust region's denser search (1,000 candidates in its box against 200 in the whole box). Fitting the trust region's GP only to the points inside it, which we expected to follow the bend, lost 1.8–2.0 decades, and 2.7 on a quadratic control too: fewer points cost more than the local shape gained (E12).

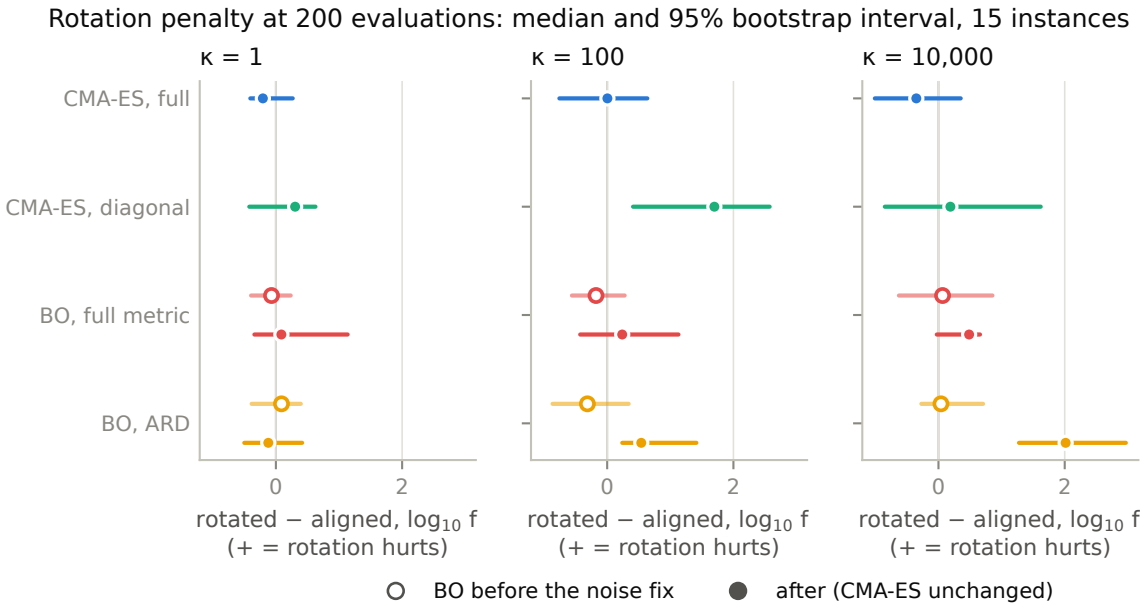


Figure 3. Part B. Rotation penalty at 200 evaluations (median, 95% bootstrap interval, 15 instances). Hollow: BO before the noise fix. The axis-restricted variant pays in both families once BO is no longer stalled.

3.6 CMA-ES keeps its invariances when set up as the theory requires; BO keeps neither (E3)

CMA-ES is designed to be invariant to strictly increasing transformations of f (it uses only rankings) and to invertible linear maps of the search space, provided its initial mean and covariance are mapped too [7]; BO has neither property by design. We tested both on the aligned $\kappa = 100$ quadratic, scoring every run on the original f and pairing it with an unchanged baseline.

Monotone transformations of f. Under $\log(1 + f)$, f^3, $\sqrt{f}$ and $f + 1$, both CMA-ES variants produced runs bit-identical to the baseline in all 120 cases (their one use of an f value, the stopping target, was set to 0, which no run reaches). BO was not invariant, and the direction depends on the transformation: cubing f cost 1.7 decades ($+1.72$ [$+1.14, +2.22$] for ARD), while $\log(1+f)$ and $f+1$ helped BO-full (-0.52, -0.68). We had expected BO's log transform to cancel powers of f; it does not, because the transform's floor is relative to the spread of observed values, and cubing stretches that spread until the floor sits at $f \approx 14$, below which BO sees no differences.

Changes of units. Measuring each coordinate in different units ($u_i = x_i/s_i$) changes the effective conditioning: $s = (1, 10)$ turns $\kappa = 100$ into 10^4, and $s = (10, 1)$ into 1. The box stays $[-5, 5]^2$ in the new units, as a user who changed nothing else would leave it. With default set-ups, no method is invariant. The harder scaling costs CMA-ES about 3.9 decades and BO 1.3–1.4. The easier one helps diagonal CMA-ES (-1.04 [-1.69, -0.07]) but hurts BO-full ($+0.91$ [$+0.62, +1.47$]; ARD $+0.45$, not significant). In the original units this scaling leaves the bowl unchanged but widens BO's box tenfold along x_1, so its uniform candidates land in the improving region less often (Section 3.4); with 200 candidates added around the incumbent the penalty's estimate falls to -0.38, though with a wide interval (E3b), so this is the likely cause. Mapping CMA-ES's initial covariance as the theory requires ($C_0 = S^{-2}$) restores invariance: to rounding error (10^{-12}) for the diagonal variant, and in distribution for the full one ($+0.21$ [-1.07, $+0.85$]). BO's equivalent, scaling each input by its own range, is invariant by construction but was not run (our box has one width). For BO the units themselves do not matter, only the valley's size relative to the box, which a change of units alters when the box's numbers stay the same. In both families, invariance to units is supplied by the set-up, not guaranteed by the algorithm.

3.7 At the box boundary, BO is drawn to the edge and CMA-ES ignores it (E4, E11)

We treat the box $[-5, 5]^2$ as a constraint. BO searches only inside it; our CMA-ES has no box handling, so we ran it as is and with a reflection back into the box (a simplified version of pycma's default). Since every method starts at the centre, each edge position (0.1 inside an edge, on it, in a corner) is compared with an interior optimum at the same distance (2-D, aligned, $\kappa \in \{1, 100\}$, 15 instances; Fig. 4).

BO is drawn to the edge. It does worse when the optimum lies 0.1 inside an edge ($+1.62$ [$+0.57, +2.10$] at $\kappa = 1$, $+0.73$ [$+0.31, +0.99$] at $\kappa = 100$), better when it lies exactly on it (-0.95, -1.68), and with the optimum in a corner it evaluates it exactly in 93–100% of runs. It places 8.5–14.5% of its evaluations exactly on the edge, a known tendency of GP-based BO [16]. Logging every step (E4b) shows that these are exploratory moves stopped by the box. EI picks a candidate that is predicted to be good and is among the most uncertain (median rank 0.84–1.0 of 200), and refining it moves it further from the data, where the uncertainty keeps rising, until the box wall stops it 0.3–0.6 units out. The edge is no more uncertain than the interior; it is simply where these moves end. Near a boundary, BO's results

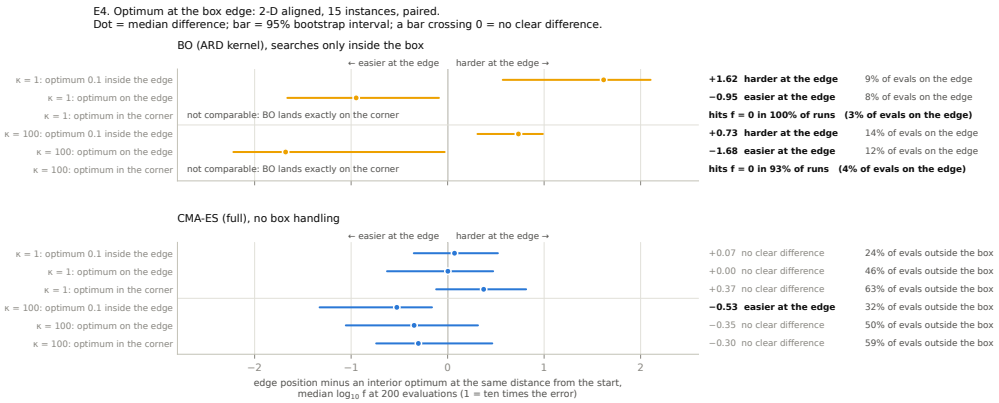


Figure 4. Part B, E4. Each edge position minus an interior optimum at the same distance from the start, median $\log_{10} f$ at 200 evaluations over 15 paired instances; bar = 95% bootstrap interval. Right: verdict, and the share of evaluations on the edge (BO) or outside the box (CMA-ES).

therefore reflect this bias, not better or worse search. It also traps BO when it searches near its best point: 4–8 of 15 runs per aligned, ill-conditioned problem stall at the edge, although about two random candidates per iteration are truly better (the GP rates their EI near 10^{-34}, for reasons not established). Reflecting candidates instead of clipping them removes the trap for the trust region on aligned problems (6 and 4 stuck runs to 0), though the axis-aligned trust region gained 3 on the rotated one, but not for local candidates (E11).

CMA-ES's accuracy is set by its distance from the start, not by the edge: no position differs clearly from its interior control, except one where the edge helps (-0.53 at $\kappa = 100$). But it ignores the box, placing 24–32% of its evaluations outside it when the optimum lies just inside an edge and 46–63% when it lies on an edge or in a corner. If the box is a hard constraint, those evaluations are not allowed. Reflection removes every violation with no significant cost in nine of ten settings; the exception is an interior optimum at $\kappa = 1$ ($+1.04$ $[+0.43, +1.64]$), where the periodic reflection lets the mean drift onto a mirror image of the optimum.

3.8 Under noise, spend the budget on new points and report the model's estimate (E5, E6, E6b)

We added Gaussian noise $\sigma \in \{0.01, 0.1, 1\}$ to the aligned $\kappa = 100$ quadratic (200 noisy evaluations, repeats included; reported points scored on the noise-free f). Repeating each point four times made the reported point worse for both families (CMA-ES by 1.1–3.0 decades, BO by 1.6 and 1.1 at $\sigma = 0.01$ and 0.1): it halves the noise but leaves a quarter as many points. Keeping 40 evaluations to re-check the best points never helped; only a larger CMA-ES population paid, at $\sigma = 1$ (-0.57 [-0.85, -0.08]). Reporting the model's estimate instead of the lowest reading, often a lucky draw, helped BO by 0.30 and 1.08 decades at $\sigma = 0.1$ and 1. With averaged repeats it hurt ($+0.89$ $[+0.00, +1.54]$, borderline), because the recommending GP learns its noise level and smooths over its best points as in Section 3.3 (E6b). Fixing BO's noise still beat fitting it under noise (-0.41 [-0.73, -0.10]; E5). With each family's best settings, BO and CMA-ES end level.

3.9 Limitations

- **Scope.** All runs are 2-D, on ellipsoids, Rosenbrock and one noisy quadratic, with 15–30 instances, so dimension effects and other landscapes are untested and several intervals are wide. Aligned and rotated instances have different optima, so rotation penalties are unpaired.
- **Implementation and mechanisms.** BO is our own NumPy/SciPy code: its log transform, relative floor and 200 uniform candidates shape the results, fixing its noise assumes a deterministic objective, and long runs reproduce only in distribution. Some explanations are hypotheses or only partly tested (the units penalty, E3b; a non-stationary kernel as a fix), and the noise results cover constant Gaussian noise only.

3.10 Conclusion

Whether a method's model can tilt decides success on rotated, ill-conditioned valleys in both families: diagonal CMA-ES, the ARD kernel and TuRBO's axis-aligned box fail where their tilting counterparts do not, and tilting costs little when it is not needed (on aligned $\kappa = 100$ the axis-aligned box was 0.57 decades better; E10). For BO this showed only after a fitting failure, a GP that called its best points noise, was found and fixed. CMA-ES keeps its rank invariance exactly and its unit invariance when its start is mapped; BO keeps neither, and its behaviour at the box edge and under noise depends on implementation choices.

4 Synthesis and conclusion

Shape and scale both decide success, within each family. Part A shows that a model with the wrong *scale* fails to find or to refine a narrow basin, and Part B that a model restricted in *shape* fails on tilted valleys in both families.

In neither part does the family alone predict the outcome: diagonal and full CMA-ES differ by up to 100 percentage points on the same rotated problem, and capped and default BO by 43 successes out of 50 on the same landscape (44 against 1 on fresh seeds).

BO's decisive fixes were to the model fit, not the acquisition. Independently, both parts found BO failing because its GP was mis-fitted, and both found acquisition changes nearly powerless: in Part A the lengthscale ran to its bound (capping it took success from 5 to 39/50; ξ, LCB and Thompson sampling at most 17/50), in Part B the fitted noise absorbed the best points (fixing it gained over a decade; LogEI nothing). In both, BO's results depend on implementation choices that published defaults leave open.

CMA-ES adapts shape well but must commit to a scale. One step size cannot both find a distant basin and refine a narrow one (Part A), but its covariance follows a rotated valley reliably (Part B); population size and restarts, which change only how much is sampled, did not move success on the primary landscape.

Trade-offs. Each fix has a price: bounding BO's lengthscale costs accuracy on the sphere, tilting costs little on aligned problems (the axis-aligned box was 0.57 decades better; E10), and BO is far more expensive per step than CMA-ES (GP fitting is cubic in the number of points, and its runs take minutes in Part B).

Scope. All headline results are 2-D, on few landscapes and 15–50 seeds, and BO runs reproduce only in distribution; the per-part limitations (Sections 2.3 and 3.9) apply.

Conclusion. Across both parts, the property that decided the outcome was whether the method's model could match the landscape's shape and scale, and this mattered more than the choice of family: each family succeeded or failed according to its own model, not as a family. In practice, before tuning BO's acquisition function, check that its GP is well fitted (lengthscale not stuck at its bound, noise not absorbing the best points); for CMA-ES, check that the covariance can tilt and that the step size suits the feature to be found and refined. These conclusions come from 2-D problems and should be re-tested in higher dimension.

References

[1] S. Ament, S. Daulton, D. Eriksson, M. Balandat, and E. Bakshy. Unexpected improvements to expected improvement for Bayesian optimization. In *Advances in Neural Information Processing Systems 36*, 2023.

[2] A. Auger and N. Hansen. A restart CMA evolution strategy with increasing population size. In *IEEE Congress on Evolutionary Computation (CEC)*, pages 1769–1776, 2005. doi: 10.1109/CEC.2005.1554902.

[3] M. Balandat, B. Karrer, D. R. Jiang, S. Daulton, B. Letham, A. G. Wilson, and E. Bakshy. BoTorch: A framework for efficient Monte-Carlo Bayesian optimization. In *Advances in Neural Information Processing Systems 33*, 2020.

[4] D. Eriksson, M. Pearce, J. Gardner, R. D. Turner, and M. Poloczek. Scalable global optimization via local Bayesian optimization. In *Advances in Neural Information Processing Systems 32*, 2019.

[5] P. I. Frazier. A tutorial on Bayesian optimization. *arXiv preprint arXiv:1807.02811*, 2018.

[6] N. Hansen. Benchmarking a BI-population CMA-ES on the BBOB-2009 function testbed. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO) Companion*, pages 2389–2396, 2009. doi: 10.1145/1570256.1570333.

[7] N. Hansen. The CMA evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.

[8] N. Hansen, S. Finck, R. Ros, and A. Auger. Real-parameter black-box optimization benchmarking 2009: Noiseless functions definitions. Technical Report RR-6829, INRIA, 2009.

[9] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. COCO: A platform for comparing continuous optimizers in a black-box setting. *Optimization Methods and Software*, 36(1):114–144, 2021.

[10] C. Hvarfner, E. O. Hellsten, and L. Nardi. Vanilla Bayesian optimization performs great in high dimensions. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

[11] D. R. Jones, M. Schonlau, and W. J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4):455–492, 1998. doi: 10.1023/A:1008306431147.

[12] T. Karvonen and C. J. Oates. Maximum likelihood estimation in Gaussian process regression is ill-posed. *Journal of Machine Learning Research*, 24(120):1–47, 2023. URL https://www.jmlr.org/papers/v24/22-1153.html.

[13] R. G. Newcombe. Interval estimation for the difference between independent proportions: comparison of eleven methods. *Statistics in Medicine*, 17(8):873–890, 1998.

[14] R. G. Newcombe. Two-sided confidence intervals for the single proportion: comparison of seven methods. *Statistics in Medicine*, 17(8):857–872, 1998.

[15] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[16] E. Siivola, A. Vehtari, J. Vanhatalo, J. González, and M. R. Andersen. Correcting boundary over-exploration deficiencies in Bayesian optimization with virtual derivative sign observations. In *IEEE 28th International Workshop on Machine Learning for Signal Processing (MLSP)*, 2018.